Congruences of Partitions: Rank Differences and Cubic Partition Pairs

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Note on the revised edition

The April 2018 report surveyed partition rank inequalities and cubic partition-pair congruences. It reproduced a partial proof of Mao's M_2-rank inequality and described an unsuccessful attempt to extend it to the progression $9m + 8$.

This edition adds a complete elementary proof in Section 4.3, including an explicit lower bound. Its inputs are the generating-function and theta-function identities already used in the original report. No claim of priority is made for the elementary argument or its lower bound.

All six chapters have been edited for mathematical notation, exposition, cross-references, and bibliographic accuracy. The discussion of cubic partition pairs remains an exposition of previously published work. Chapter 6 also records Chern's earlier resolution of Lin's conjectures.

The original report, including its title page and supervisor's certificate, is preserved without alteration as `report-2018.pdf`. The authorship, supervision, and certification recorded there refer to the 2018 submission; this dated revision is a subsequent edition. A separate four-page note, `mao-rank-inequality.pdf`, presents the elementary proof on its own.

Abstract

This report studies partition rank inequalities and congruences for cubic partition pairs. After introducing partitions, generating functions, and basic q-series identities, we discuss Dyson's rank and the M_2-rank of partitions without repeated odd parts.

The principal addition in this revised edition is an elementary completion of the approach of Barman and Sachdeva to Mao's rank inequality. Writing

$$d(n) = N_2(0, 6, n) + N_2(1, 6, n) - N_2(2, 6, n) - N_2(3, 6, n),$$

we prove

$$d(3n + 2) \geq p_S(\lfloor n/3 \rfloor) \geq 1 \qquad (n \geq 0),$$

where $p_S(m)$ counts partitions of m into parts not congruent to 2 modulo 4. The proof pairs consecutive terms of an alternating Lambert series and retains the partition-generating product needed for coefficientwise positivity. It includes the previously untreated progression $d(9m + 8)$.

The second part surveys cubic partition pairs and the congruences conjectured by Lin, presenting the deductions from the work of Gireesh and Mahadeva Naika. These results are distinguished from the 2026 addition to the rank-inequality argument.

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Part I

Partition rank inequalities

Chapter 1

Introduction to partitions

1.1 Background

A partition of a nonnegative integer n is a nonincreasing sequence of positive integers whose sum is n. Two sums differing only in the order of their summands represent the same partition. The notation $\lambda \vdash n$ means that λ is a partition of n. The empty partition is the unique partition of zero.

A summand is called a part. For a nonempty partition λ, write $n(\lambda)$ for its number of parts and $l(\lambda)$ for its largest part. For example, the five partitions of 4 are

$$(4), \quad (3,1), \quad (2,2), \quad (2,1,1), \quad (1,1,1,1).$$

Thus $n((2,1,1)) = 3$ and $l((2,1,1)) = 2$.

1.2 Definitions and terminology

1.2.1 The q-Pochhammer symbol

The q-Pochhammer symbol is defined by

$$(a;q)_0 = 1, \qquad (a;q)_n = \prod_{k=0}^{n-1}(1-aq^k) \quad (n \geq 1),$$

and

$$(a;q)_\infty = \prod_{k=0}^{\infty}(1-aq^k) \qquad (|q| < 1).$$

When the base is clear, we abbreviate $(a;q)_n$ to $(a)_n$. The following product identities will be useful:

$$(q^a;q^b)_\infty(-q^a;q^b)_\infty = (q^{2a};q^{2b})_\infty, \tag{1.1}$$

$$(cq^a;q^{2b})_\infty(cq^{a+b};q^{2b})_\infty = (cq^a;q^b)_\infty. \tag{1.2}$$

Here the parameters are chosen so that the products converge. For the partition-generating products below, the same identities hold formally. We also use the notation

$$\begin{aligned}(a_1,\dots,a_k;q)_n &= \prod_{i=1}^{k}(a_i;q)_n,\\ J_b &= (q^b;q^b)_\infty,\\ J_{a,b} &= (q^a,q^{b-a},q^b;q^b)_\infty \quad (0<a<b),\\ \chi(q) &= (-q;q^2)_\infty.\end{aligned}$$

1.2.2 Ramanujan's general theta function

Ramanujan's general theta function is

$$f(a,b) = \sum_{n\in\mathbb{Z}} a^{n(n+1)/2}b^{n(n-1)/2}, \qquad |ab|<1. \tag{1.3}$$

The specializations used in this report are

$$\begin{aligned}\varphi(q) = f(q,q) &= \sum_{n\in\mathbb{Z}} q^{n^2} = (-q,-q,q^2;q^2)_\infty,\\ \psi(q) = f(q,q^3) &= \sum_{n\geq 0} q^{n(n+1)/2} = \frac{(q^2;q^2)_\infty}{(q;q^2)_\infty}.\end{aligned}$$

These product evaluations follow from the Jacobi triple product in Chapter 2; see also Berndt [6, 5].

1.2.3 Partition functions

Let $p(n)$ denote the number of partitions of n, with $p(0)=1$. Write $p_m(n)$ for the number of partitions whose parts are at most m. For a set S of positive integers, let $p(S,m,n)$ count partitions of n into exactly m parts from S. When S is the set of all positive integers, abbreviate this to $p(m,n)$.

Similarly, $Q(n)$ counts partitions into distinct parts, and $Q(S, m, n)$ counts partitions of n into exactly m distinct parts from S. We use $Q(m, n)$ when S is unrestricted, and set $Q(0) = 1$.

1.2.4 Overpartitions

An overpartition is a partition in which the first occurrence of each distinct part may be overlined. The eight overpartitions of 3 are

$$(3),\ (\overline{3}),\ (2,1),\ (2,\overline{1}),\ (\overline{2},1),\ (\overline{2},\overline{1}),\ (1,1,1),\ (\overline{1},1,1).$$

Writing $\overline{p}(n)$ for their number, we have

$$\overline{p}(n) = \sum_{k=0}^{n} p(n-k)Q(k).$$

Indeed, the overlined parts form a partition into distinct parts, while the non-overlined parts form an unrestricted partition.

1.3 Generating functions

The generating function of a sequence $(a_n)_{n\geq 0}$ is the formal power series $\sum_{n\geq 0} a_n q^n$. For ordinary partitions, choosing the multiplicity of each part independently gives

$$\sum_{n\geq 0} p(n)q^n = \prod_{k\geq 1}(1 + q^k + q^{2k} + \cdots) = \frac{1}{(q;q)_\infty}. \tag{1.4}$$

More generally,

$$\sum_{n,m\geq 0} p(S,m,n)z^m q^n = \prod_{k\in S} \frac{1}{1-zq^k},$$
$$\sum_{n,m\geq 0} Q(S,m,n)z^m q^n = \prod_{k\in S}(1+zq^k).$$

The convolution for overpartitions therefore gives

$$\sum_{n\geq 0} \overline{p}(n)q^n = \frac{(-q;q)_\infty}{(q;q)_\infty}.$$

All coefficient extractions in this report are formal. Analytic identities are initially stated in their domains of convergence. Congruences between series with integer coefficients are understood coefficientwise.

Chapter 2

Fundamental properties of q-series and theta functions

2.1 The q-binomial theorem

The q-binomial theorem generalizes the binomial expansion: replacing a by q^α and taking $q \to 1^-$ recovers the series for $(1-z)^{-\alpha}$.

Theorem 2.1.1. *For $|q| < 1$ and $|z| < 1$,*

$$\sum_{n\geq 0} \frac{(a;q)_n}{(q;q)_n} z^n = \frac{(az;q)_\infty}{(z;q)_\infty}. \tag{2.1}$$

Proof. The product on the right converges uniformly on compact subsets of $|z| < 1$ and defines an analytic function there. Write

$$F(z) = \frac{(az;q)_\infty}{(z;q)_\infty} = \sum_{n\geq 0} A_n z^n.$$

The product representation gives $(1-z)F(z) = (1-az)F(qz)$. Comparing coefficients for $n \geq 1$ yields

$$A_n - A_{n-1} = q^n A_n - aq^{n-1} A_{n-1}, \qquad A_n = \frac{1-aq^{n-1}}{1-q^n} A_{n-1}.$$

Since $A_0 = 1$, induction gives $A_n = (a;q)_n/(q;q)_n$. □

2.1.1 Euler's identities

Two useful consequences are

$$\sum_{n\geq 0} \frac{z^n}{(q;q)_n} = \frac{1}{(z;q)_\infty} \qquad (|z|<1), \tag{2.2}$$

$$\sum_{n\geq 0} \frac{(-z)^n q^{n(n-1)/2}}{(q;q)_n} = (z;q)_\infty. \tag{2.3}$$

The first follows by setting $a = 0$ in (2.1). For the second, replace z by z/a and let $a \to \infty$; the resulting identity holds for all complex z when $|q| < 1$.

2.2 The Jacobi triple product identity

Theorem 2.2.1. *For $z \neq 0$ and $0 < |q| < 1$,*

$$\sum_{n\in\mathbb{Z}} z^n q^{n^2} = (-zq;q^2)_\infty (-q/z;q^2)_\infty (q^2;q^2)_\infty. \tag{2.4}$$

Proof. Applying (2.3) with base q^2 gives

$$\begin{aligned}(-zq;q^2)_\infty &= \sum_{n\geq 0} \frac{z^n q^{n^2}}{(q^2;q^2)_n} \\ &= \frac{1}{(q^2;q^2)_\infty} \sum_{n\in\mathbb{Z}} z^n q^{n^2} (q^{2n+2};q^2)_\infty.\end{aligned}$$

The extension to negative n is valid because the product in each such term contains the factor $1 - q^0 = 0$. Expand that product using (2.3). Initially assuming $|q/z| < 1$, absolute convergence permits rearrangement, giving

$$\begin{aligned}(-zq;q^2)_\infty &= \frac{1}{(q^2;q^2)_\infty} \sum_{n\in\mathbb{Z}} \sum_{r\geq 0} \frac{z^n (-1)^r q^{n^2+(2n+2)r+r(r-1)}}{(q^2;q^2)_r} \\ &= \frac{1}{(q^2;q^2)_\infty} \sum_{r\geq 0} \frac{(-q/z)^r}{(q^2;q^2)_r} \sum_{m\in\mathbb{Z}} z^m q^{m^2} \\ &= \frac{\sum_{m\in\mathbb{Z}} z^m q^{m^2}}{(q^2;q^2)_\infty (-q/z;q^2)_\infty}.\end{aligned}$$

In the second line we set $m = n + r$; the third line uses (2.2). Rearranging proves (2.4) for $|z| > |q|$. Both sides of (2.4) are analytic for $z \neq 0$, so analytic continuation proves the identity throughout that domain. □

2.2.1 Euler's pentagonal number theorem

A specialization of (2.4) gives

$$(q;q)_\infty = \sum_{n\in\mathbb{Z}} (-1)^n q^{n(3n-1)/2} = \sum_{n\in\mathbb{Z}} (-1)^n q^{n(3n+1)/2}. \tag{2.5}$$

The two sums agree after replacing n by $-n$. These identities are also valid as formal power series.

Chapter 3

Ranks

3.1 Ramanujan's and Dyson's work

Ramanujan's classical congruences state that, for every $k \geq 0$,

$$p(5k+4) \equiv 0 \pmod 5, \tag{3.1}$$

$$p(7k+5) \equiv 0 \pmod 7, \tag{3.2}$$

$$p(11k+6) \equiv 0 \pmod{11}. \tag{3.3}$$

We give a proof of (3.1); see Berndt [6] for further background on these congruences.

Proof of (3.1). Put $E = (q;q)_\infty$ and $J = E^3$. Euler's pentagonal number theorem and Jacobi's identity for the cube of the Euler product give

$$E = \sum_{n\in\mathbb{Z}} (-1)^n q^{n(3n+1)/2}, \qquad J = \sum_{n\geq 0} (-1)^n (2n+1) q^{n(n+1)/2}.$$

The exponents in E are congruent to 0, 1, or 2 modulo 5. Those in J are congruent to 0, 1, or 3; in the last case $n \equiv 2 \pmod 5$, so $5 \mid 2n+1$. If E_i and J_i denote the corresponding residue-class subseries, then

$$E = E_0 + E_1 + E_2, \qquad J \equiv J_0 + J_1 \pmod 5.$$

Furthermore, the binomial theorem gives $E^5 \equiv (q^5;q^5)_\infty \pmod 5$. Thus

$$\sum_{n\geq 0} p(n) q^n = \frac{EJ}{E^5} \equiv \frac{(E_0+E_1+E_2)(J_0+J_1)}{(q^5;q^5)_\infty} \pmod 5.$$

The numerator has no terms whose exponents are 4 modulo 5, and the reciprocal denominator contains only multiples of 5. Comparing coefficients proves the claim. □

The first two congruences also follow immediately from the identities

$$\sum_{k\geq 0} p(5k+4)q^k = 5\frac{(q^5;q^5)_\infty^5}{(q;q)_\infty^6}, \tag{3.4}$$

$$\sum_{k\geq 0} p(7k+5)q^k = 7\frac{(q^7;q^7)_\infty^3}{(q;q)_\infty^4} + 49q\frac{(q^7;q^7)_\infty^7}{(q;q)_\infty^8}. \tag{3.5}$$

These are recorded here as known identities [6, 5]; the proof above does not by itself establish the congruences modulo 7 or 11.

3.1.1 The rank of a partition

Dyson [12] defined the rank of a nonempty partition by

$$r(\lambda) = l(\lambda) - n(\lambda).$$

For its conjugate partition λ^*, we have $r(\lambda^*) = -r(\lambda)$. In particular, the rank distribution is symmetric. The empty partition is assigned rank zero.

Write $N(s,n)$ for the number of partitions of n with rank s, and $N(s,m,n)$ for the number whose rank is s modulo m. Then

$$N(s,m,n) = \sum_{k\in\mathbb{Z}} N(mk+s,n).$$

Dyson conjectured the equalities

$$N(s,5,5n+4) = \frac{p(5n+4)}{5}, \qquad N(t,7,7n+5) = \frac{p(7n+5)}{7}$$

for every residue s modulo 5, every residue t modulo 7, and $n \geq 0$. These equalities divide the relevant partitions into equally sized classes, giving combinatorial interpretations of Ramanujan's first two congruences.

Atkin and Swinnerton-Dyer [2] proved Dyson's conjectures by finding generating functions for rank differences

$$N(s,\ell,\ell n+d) - N(t,\ell,\ell n+d)$$

with $\ell = 5, 7$ and $0 \leq s,t,d < \ell$. The differences vanish for $(\ell,d) = (5,4)$ and $(7,5)$.

3.1.2 The M_2-rank

For a partition λ without repeated odd parts, define

$$r_2(\lambda) = \left\lceil \frac{l(\lambda)}{2} \right\rceil - n(\lambda).$$

Even parts may be repeated. Again, the empty partition has rank zero. Let $N_2(s, n)$ count these partitions with M_2-rank equal to s, and let $N_2(s, m, n)$ count those with rank congruent to s modulo m. Differences between these counts are the subject of the next chapter.

Chapter 4

Mao's inequalities: partial results and an elementary completion

4.1 The conjectures and their status

Mao [20, 21] derived generating functions for Dyson rank differences modulo 10 and M_2-rank differences modulo 6 and 10. Among the inequalities he proved is

$$N_2(0,6,3n+1) + N_2(1,6,3n+1) > N_2(2,6,3n+1) + N_2(3,6,3n+1) \qquad (n \geq 0).$$

The following statements were originally conjectured by Mao; they are listed in the form discussed by Barman and Sachdeva [3].

$$N(0,10,5n) + N(1,10,5n) > N(4,10,5n) + N(5,10,5n), \tag{4.1}$$

$$N(1,10,5n) + N(2,10,5n) \geq N(3,10,5n) + N(4,10,5n), \tag{4.2}$$

$$N_2(0,10,5n) + N_2(1,10,5n) > N_2(4,10,5n) + N_2(5,10,5n), \tag{4.3}$$

$$N_2(0,10,5n+4) + N_2(1,10,5n+4) > N_2(4,10,5n+4) + N_2(5,10,5n+4), \tag{4.4}$$

$$N_2(1,10,5n) + N_2(2,10,5n) > N_2(3,10,5n) + N_2(4,10,5n), \tag{4.5}$$

$$N_2(1,10,5n+2) + N_2(2,10,5n+2) > N_2(3,10,5n+2) + N_2(4,10,5n+2), \tag{4.6}$$

$$N_2(0,6,3n+2) + N_2(1,6,3n+2) > N_2(2,6,3n+2) + N_2(3,6,3n+2). \tag{4.7}$$

Here $n \geq 1$ in (4.2), (4.5), and (4.6), and $n \geq 0$ in the other statements.

Alwaise, Iannuzzi, and Swisher [1] proved (4.1)–(4.4), including the strict version of (4.2) for $n \geq 1$. Barman and Sachdeva [3] proved (4.7) for two of the three residue classes of n modulo 3. The original project investigated the remaining class.

Jennings-Shaffer and Reihill [15] subsequently resolved the remaining inequalities using asymptotic formulas. Section 4.3 gives an elementary proof of (4.7) using the generating-function approach of Barman and Sachdeva.

4.2 The partial result of Barman and Sachdeva

For the rest of the chapter, put

$$d(n) = N_2(0,6,n) + N_2(1,6,n) - N_2(2,6,n) - N_2(3,6,n). \tag{4.8}$$

Theorem 4.2.1 (Barman–Sachdeva)**.** *For all* $m \geq 0$, $d(9m+2) > 0$ *and* $d(9m+5) > 0$. *Equivalently,* $d(3n+2) > 0$ *when* $3 \nmid n+1$.

The proof uses the following identity of Mao, reproduced in [3].

Theorem 4.2.2 (Mao)**.**

$$\begin{aligned}\sum_{n\geq 0} d(n)q^n = \frac{1}{J_{9,36}} \sum_{j\in\mathbb{Z}} \frac{(-1)^j q^{18j^2+9j}}{1-q^{18j+3}} + q\frac{J_{6,36}^2 J_{18,36} J_{36}^3}{J_{3,36}^2 J_{9,36} J_{15,36}^2} \\ + \frac{J_{6,36} J_{18,36}^2 J_{36}^3}{2qJ_{9,36} J_{3,36}^2 J_{15,36}^2} - \frac{1}{J_{9,36}} \sum_{j\in\mathbb{Z}} \frac{(-1)^j q^{18j^2+9j-1}}{1+q^{18j}}.\end{aligned} \tag{4.9}$$

Negative-index summands are rewritten with nonnegative denominator exponents before their formal expansion. The apparent negative powers in the last two terms cancel in their sum.

The theta identity needed below is due to Baruah and Barman [4]:

$$\varphi(q)^2 + \varphi(q^3)^2 = 2\varphi(-q^6)^2 \frac{\chi(q)\psi(-q^3)}{\chi(-q)\psi(q^3)}. \tag{4.10}$$

Proposition 4.2.3 (Barman–Sachdeva)**.**

$$\sum_{n\geq 0} d(3n+2)q^n = \frac{1}{qJ_{3,12}} \left(\frac{J_{2,12} J_{6,12}^2 J_{12}^3}{2J_{1,12}^2 J_{5,12}^2} - \sum_{j\in\mathbb{Z}} \frac{(-1)^j q^{6j^2+3j}}{1+q^{6j}} \right). \tag{4.11}$$

Proof. In (4.9), the first term contains only exponents divisible by 3, the second only exponents congruent to 1, and the sum of the last two only exponents congruent to 2. Retain the last two terms, divide by q^2, and replace q^3 by q. This gives (4.11). □

Lemma 4.2.4 (Barman–Sachdeva)**.**

$$\frac{J_{2,12} J_{6,12}^2 J_{12}^3}{J_{1,12}^2 J_{5,12}^2} = \frac{\varphi(q)^2 + \varphi(q^3)^2}{2}. \tag{4.12}$$

Proof. Expand the products and apply (1.1) and (1.2). The expression on the left becomes

$$\varphi(-q^6)^2\frac{(-q,-q^5;q^6)_\infty}{(q,q^5;q^6)_\infty}=\varphi(-q^6)^2\frac{\chi(q)\psi(-q^3)}{\chi(-q)\psi(q^3)}.$$

Equation (4.10) completes the proof. □

Proof of Theorem 4.2.1. Combining (4.11) and (4.12) gives

$$D(q):=\sum_{n\geq 0}d(3n+2)q^n=\frac{1}{qJ_{3,12}}\left(\frac{\varphi(q)^2+\varphi(q^3)^2}{4}-\sum_{j\in\mathbb{Z}}\frac{(-1)^jq^{6j^2+3j}}{1+q^{6j}}\right). \tag{4.13}$$

Both $1/J_{3,12}$ and the bilateral sum contain only exponents divisible by 3. Therefore, if $3\nmid n+1$,

$$d(3n+2)=[q^{n+1}]\frac{\varphi(q)^2+\varphi(q^3)^2}{4J_{3,12}}.$$

Here $[q^k]F(q)$ denotes the coefficient of q^k in $F(q)$. The numerator divided by 4 has nonnegative coefficients and contains $q+q^2$. The reciprocal product has nonnegative coefficients and contains $1/(1-q^3)$ as a factor. Thus the coefficient above is positive whenever $n+1\equiv 1$ or $2 \pmod 3$. □

4.2.1 The obstruction in the original attempt

The preceding proof does not address $n\equiv 2 \pmod 3$, corresponding to $d(9m+8)$. The original attempt compared coefficients of the theta series with those of the alternating bilateral sum. The numerator in (4.13) is not coefficientwise nonnegative: its coefficient of q^{27} is -1.

Barman and Sachdeva also suggested testing positivity after dividing that numerator by $1-q^{12}$. Such positivity would be a useful sufficient step, but it is not an equivalent reformulation of the desired positivity after multiplication by the full reciprocal product. The argument below keeps that product and uses its factors in the cancellation.

4.3 An elementary completion (2026 addition)

For formal power series, write $A\succeq B$ when every coefficient of $A-B$ is nonnegative. Define

$$S=\{k\geq 1: k\not\equiv 2 \pmod 4\},\qquad P(q)=\prod_{k\in S}\frac{1}{1-q^k}=\sum_{m\geq 0}p_S(m)q^m. \tag{4.14}$$

Then $p_S(m)$ counts partitions into odd parts and multiples of four, and $1/J_{3,12}=P(q^3)$.

Theorem 4.3.1. *For every $n \geq 0$,*

$$d(3n+2) \geq p_S(\lfloor n/3 \rfloor) \geq 1. \tag{4.15}$$

In particular, $d(9m+a) \geq p_S(m)$ for $m \geq 0$ and $a \in \{2,5,8\}$.

Pairing the terms with indices j and $-j$ in (4.13), we obtain

$$\sum_{j\in\mathbb{Z}} \frac{(-1)^j q^{6j^2+3j}}{1+q^{6j}} = \frac{1}{2} + 2\sum_{j\geq 1} \frac{(-1)^j q^{6j^2+3j}}{1+q^{6j}}.$$

Indeed, multiplying numerator and denominator of the negative-index term by q^{6j} makes it equal to the positive-index term. Put

$$T(q) = \frac{\varphi(q)^2 + \varphi(q^3)^2 - 2}{4}, \qquad L(q) = \sum_{j\geq 1} \frac{(-1)^{j+1} q^{2j^2+j}}{1+q^{2j}}. \tag{4.16}$$

Equation (4.13) becomes

$$D(q) = \frac{1}{q}\left(P(q^3)T(q) + 2P(q^3)L(q^3)\right). \tag{4.17}$$

The bracket has zero constant term, so division by q is valid formally.

Lemma 4.3.2. *We have $P(q)L(q) \succeq 0$.*

Proof. For each positive odd integer r, pair the term of index r with the term of index $r+1$. Their contribution after multiplication by $P(q)$ is

$$B_r(q) = P(q)\left(\frac{q^{2r^2+r}}{1+q^{2r}} - \frac{q^{2(r+1)^2+r+1}}{1+q^{2r+2}}\right).$$

The difference between the numerator exponents is $4r+3$. Therefore

$$B_r(q) = q^{2r^2+r}P(q)\frac{(1-q^{4r+3}) + q^{2r+2}(1-q^{4r+1})}{(1+q^{2r})(1+q^{2r+2})}. \tag{4.18}$$

Use the factorization

$$\frac{1}{(1+q^{2r})(1+q^{2r+2})} = \frac{(1-q^r)(1+q^r)(1-q^{2r+2})}{(1-q^{4r})(1-q^{4r+4})}.$$

For $s = 4r + 1$ or $4r + 3$, define

$$\begin{aligned} Q_{r,s}(q) &= P(q)(1 - q^r)(1 - q^{2r+2})(1 - q^s) \\ &= \prod_{k \in S \setminus \{r, 2r+2, s\}} \frac{1}{1 - q^k} \succeq 0. \end{aligned}$$

The cancellation is valid because r and s are odd, $2r + 2$ is divisible by four, and $r < 2r + 2 < 4r + 1 \leq s$. Thus the three cancelled factors correspond to distinct allowed part sizes. Substituting into (4.18) gives

$$B_r(q) = \frac{q^{2r^2+r}(1 + q^r)}{(1 - q^{4r})(1 - q^{4r+4})} \left(Q_{r,4r+3}(q) + q^{2r+2} Q_{r,4r+1}(q) \right) \succeq 0. \tag{4.19}$$

Summing over odd $r \geq 1$ proves the lemma. Regrouping is valid because $2r^2 + r \to \infty$, so each coefficient receives only finitely many contributions. □

Proof of Theorem 4.3.1. The series $T(q)$ has nonnegative coefficients and satisfies $T(q) \succeq q + q^2 + q^3$. The first two monomials come from $\varphi(q)^2/4$ and the third from $\varphi(q^3)^2/4$. Using Lemma 4.3.2 in (4.17) gives

$$D(q) \succeq P(q^3)(1 + q + q^2) = \sum_{m \geq 0} p_S(m)(q^{3m} + q^{3m+1} + q^{3m+2}).$$

Comparing coefficients proves the first inequality in (4.15). A partition into ones exists for every $m > 0$, and $p_S(0) = 1$, so the lower bound is always positive. □

4.3.1 Verification and scope

The script `verify_completion.py` checks the formulas against direct rank counts through $n = 300$, the generating-function identity through q^{1273}, and the paired product identities through q^{1274}. For the formerly missing progression, the first ten values are

$$(d(9m + 8))_{m=0}^{9} = (1, 2, 5, 7, 9, 15, 23, 31, 40, 56).$$

These finite checks support the algebra and indexing. The formal proof above establishes the result for all n independently of those checks.

Part II

Cubic partition pairs

Chapter 5

Cubic partition-pair congruences

5.1 Definitions

5.1.1 Cubic partitions and overcubic partitions

Chan [7, 8, 9] studied the cubic partition function $a(n)$, defined by

$$\sum_{n\geq 0} a(n)q^n = \frac{1}{(q;q)_\infty (q^2;q^2)_\infty}.$$

A cubic partition may be viewed as an ordered pair of partitions, the second containing only even parts. Kim [16] introduced the corresponding overcubic partition function

$$\sum_{n\geq 0} \overline{a}(n)q^n = \frac{(-q;q)_\infty (-q^2;q^2)_\infty}{(q;q)_\infty (q^2;q^2)_\infty}.$$

5.1.2 Cubic partition pairs

Zhao and Zhong [23] studied the function $b(n)$ defined by

$$\sum_{n\geq 0} b(n)q^n = \frac{1}{(q;q)_\infty^2 (q^2;q^2)_\infty^2}. \tag{5.1}$$

It counts ordered pairs of cubic partitions with total size n. Kim [17] studied statistics explaining congruences such as $b(5n+4) \equiv 0 \pmod 5$. The overpartition analogue [18] is

$$\sum_{n\geq 0} \overline{b}(n)q^n = \frac{(-q;q)_\infty^2 (-q^2;q^2)_\infty^2}{(q;q)_\infty^2 (q^2;q^2)_\infty^2}.$$

5.2 Preliminaries

Throughout this part, write

$$f_t = (q^t;q^t)_\infty \quad (t \geq 1), \qquad \mathcal{A}(q) = \sum_{m,n\in\mathbb{Z}} q^{m^2+mn+n^2}.$$

The symbol $\mathcal{A}(q)$ denotes a theta series, distinct from the generating function for $a(n)$. Euler's pentagonal number theorem gives

$$f_1 = \sum_{m\in\mathbb{Z}} (-1)^m q^{m(3m+1)/2}.$$

Lemma 5.2.1. *For every positive integer t,*

$$f_t^3 \equiv f_{3t} \pmod 3, \qquad f_t^9 \equiv f_{3t}^3 \pmod 9.$$

Proof. For each factor, $(1-x)^3 \equiv 1-x^3 \pmod 3$. Writing $(1-x)^3 = 1-x^3+3R(x)$ and cubing shows $(1-x)^9 \equiv (1-x^3)^3 \pmod 9$. Multiplication over the factors proves the assertions coefficientwise. □

All f_t have constant term one and are therefore invertible as formal power series. The congruences may consequently also be used in quotients.

The product evaluations of the theta functions yield

$$\varphi(-q) = \frac{f_1^2}{f_2}, \qquad \psi(q) = \frac{f_2^2}{f_1}.$$

We also use the following dissection identities [23]:

$$\frac{1}{f_1 f_2} = \mathcal{A}(q^6)\frac{f_9^3}{f_3^4 f_6^3} + q\mathcal{A}(q^3)\frac{f_{18}^3}{f_3^3 f_6^4} + 3q^2\frac{f_9^3 f_{18}^3}{f_3^4 f_6^4}, \tag{5.2}$$

$$f_1 f_2 = \frac{f_6 f_9^4}{f_3 f_{18}^2} - q f_9 f_{18} - 2q^2\frac{f_3 f_{18}^4}{f_6 f_9^2}. \tag{5.3}$$

Each product quotient multiplying 1, q, or q^2 is a series in q^3. This makes extraction of a specified residue class possible.

5.3 Lin's congruence modulo 27

Lin [19] established the generating-function congruence

$$\sum_{n\geq 0} b(9n+7)q^n \equiv 18 f_1 f_2 \left(\frac{f_3^7}{f_6} + q\frac{f_6^7}{f_3} \right) - 9 f_3^2 f_6^2 \pmod{27}. \tag{5.4}$$

We use it to deduce the following result, following Lin's argument.

Theorem 5.3.1 (Lin)**.** *For every $n \geq 0$, $b(81n+61) \equiv 0 \pmod{27}$.*

Proof. Substitute (5.3) into (5.4), retain the terms whose exponents are divisible by 3, and replace q^3 by q. This gives

$$\begin{aligned}\sum_{n\geq 0} b(27n+7)q^n &\equiv 18 \left(\frac{f_1^6 f_3^4}{f_6^2} - 2q\frac{f_2^6 f_6^4}{f_3^2} \right) - 9 f_1^2 f_2^2 \pmod{27} \\ &\equiv 18 \left(\frac{f_3^6}{f_6^2} - 2q\frac{f_6^6}{f_3^2} \right) - 9\frac{f_3 f_6}{f_1 f_2} \pmod{27}. \end{aligned} \tag{5.5}$$

In the second line we used Lemma 5.2.1 modulo 3; the prefactors 18 and 9 make this sufficient modulo 27. Substituting (5.2) into the last term yields

$$\begin{aligned}\sum_{n\geq 0} b(27n+7)q^n \equiv{}& 18 \left(\frac{f_3^6}{f_6^2} - 2q\frac{f_6^6}{f_3^2} \right) \\ &- 9 f_3 f_6 \left(\mathcal{A}(q^6)\frac{f_9^3}{f_3^4 f_6^3} + q\mathcal{A}(q^3)\frac{f_{18}^3}{f_3^3 f_6^4} \right) \pmod{27}.\end{aligned}$$

The omitted q^2 term has a coefficient divisible by 27. Every remaining term has exponent congruent to 0 or 1 modulo 3. The coefficient of q^{3n+2} on the left is $b(27(3n+2)+7) = b(81n+61)$, which proves the theorem. □

Chapter 6

Resolutions of Lin's conjectures

6.1 Lin's conjectures and subsequent proofs

Lin [19] conjectured the stronger congruence

$$b(81n+61) \equiv 0 \pmod{243} \qquad (n \geq 0) \tag{6.1}$$

and the generating-function congruences

$$\sum_{n\geq 0} b(81n+7)q^n \equiv 9\frac{f_2 f_3^2}{f_6} \pmod{81}, \tag{6.2}$$

$$\sum_{n\geq 0} b(81n+34)q^n \equiv 36\frac{f_1 f_6^2}{f_3} \pmod{81}. \tag{6.3}$$

These are now theorems. Chern [10] resolved the conjectures and proved the stronger divisibility $b(81n+61) \equiv 0 \pmod{729}$. Gireesh and Mahadeva Naika [13] subsequently gave a general family of congruences and alternative proofs of Lin's conjectures. This chapter presents the deductions used in the original project report.

6.2 The general family of congruences

Gireesh and Mahadeva Naika use identities of Cooper [11] together with the huffing operator

$$H\left(\sum_{n\geq 0} c_n q^n\right) = \sum_{n\geq 0} c_{3n} q^{3n}.$$

Their recurrence and valuation arguments establish the following theorem.

Theorem 6.2.1 (Gireesh–Mahadeva Naika, [13]). *For all integers $\alpha, n \geq 0$,*

$$b\left(3^{2\alpha+1}n + \frac{3^{2\alpha+1}+1}{4}\right) \equiv 0 \pmod{3^{2\alpha}}, \tag{6.4}$$

$$b\left(3^{2\alpha+4}n + \frac{3^{2\alpha+5}+1}{4}\right) \equiv 0 \pmod{3^{2\alpha+5}}, \tag{6.5}$$

$$b\left(3^{2\alpha+5}n + \frac{7\cdot 3^{2\alpha+4}+1}{4}\right) \equiv 0 \pmod{3^{\alpha+6}}, \tag{6.6}$$

$$b\left(3^{2\alpha+5}n + \frac{11\cdot 3^{2\alpha+4}+1}{4}\right) \equiv 0 \pmod{3^{\alpha+6}}. \tag{6.7}$$

Setting $\alpha = 0$ in (6.5) gives (6.1) immediately. We cite the general theorem rather than reproduce its recurrence argument.

6.3 The two congruences modulo 81

We give the deduction in [13, Section 5], making explicit the product identities and coefficient extractions it uses.

Lemma 6.3.1.

$$\begin{aligned}\frac{f_2^5}{f_1^4} = \frac{f_6^6 f_9^{10}}{f_3^{10} f_{18}^5} + 4q\frac{f_6^5 f_9^7}{f_3^9 f_{18}^2} + 9q^2\frac{f_6^4 f_9^4 f_{18}}{f_3^8} \\ + 10q^3\frac{f_6^3 f_9 f_{18}^4}{f_3^7} + 4q^4\frac{f_6^2 f_{18}^7}{f_3^6 f_9^2}.\end{aligned} \tag{6.8}$$

Proof. We use the two identities recorded in [13, Section 5]:

$$\frac{f_2^2}{f_1} = \frac{f_6 f_9^2}{f_3 f_{18}} + q\frac{f_{18}^2}{f_9}, \tag{6.9}$$

$$\frac{f_2}{f_1^2} = \frac{f_6^4 f_9^6}{f_3^8 f_{18}^3} + 2q\frac{f_6^3 f_9^3}{f_3^7} + 4q^2\frac{f_6^2 f_{18}^3}{f_3^6}. \tag{6.10}$$

The first is a corollary in Berndt [5, p. 49]; the second is used by Hirschhorn and Sellers [14]. Squaring (6.9) and multiplying by (6.10) gives (6.8). □

The next two inputs are the congruences recorded in [13, Section 5]:

$$U(q) := \frac{f_2^4 f_3^8}{f_1^8 f_6^4} + q\frac{f_2 f_6^5}{f_1^5 f_3} \equiv 1 \pmod 9, \tag{6.11}$$

$$\sum_{n\geq 0} b(27n+7)q^n \equiv 9\frac{f_3^2 f_6^2}{f_1^4 f_2^4} \equiv 9\frac{f_3^2 f_2^5}{f_1^4 f_6} \pmod{81}. \tag{6.12}$$

The first follows from a theta identity of Son [22]. The first congruence in (6.12) is an input from the recurrence calculation in [13]; the second uses $f_2^9 \equiv f_6^3 \pmod 9$.

Theorem 6.3.2. *The congruences* (6.2) *and* (6.3) *hold.*

Proof. Combining (6.8) and (6.12) gives

$$\sum_{n\geq 0} b(27n+7)q^n \equiv 9\frac{f_3^2}{f_6}\left(\frac{f_6^6 f_9^{10}}{f_3^{10} f_{18}^5} + 4q\frac{f_6^5 f_9^7}{f_3^9 f_{18}^2} + q^3\frac{f_6^3 f_9 f_{18}^4}{f_3^7} + 4q^4\frac{f_6^2 f_{18}^7}{f_3^6 f_9^2}\right) \pmod{81}. \tag{6.13}$$

The coefficient $9q^2$ in the dissection vanishes after multiplication by 9 modulo 81, and 10 may be replaced by 1 in the q^3 term.

Retain the terms with exponents divisible by 3 in (6.13) and replace q^3 by q. We obtain

$$\begin{aligned}\sum_{n\geq 0} b(81n+7)q^n &\equiv 9\frac{f_1^2}{f_2}\left(\frac{f_2^6 f_3^{10}}{f_1^{10} f_6^5} + q\frac{f_2^3 f_3 f_6^4}{f_1^7}\right) \pmod{81}\\ &\equiv 9\frac{f_2 f_3^2}{f_6}U(q) \pmod{81}.\end{aligned}$$

Using (6.11) proves (6.2).

Next retain the terms with exponents congruent to 1 modulo 3, divide by q, and replace q^3 by q. Then

$$\begin{aligned}\sum_{n\geq 0} b(81n+34)q^n &\equiv 36\frac{f_1^2}{f_2}\left(\frac{f_2^5 f_3^7}{f_1^9 f_6^2} + q\frac{f_2^2 f_6^7}{f_1^6 f_3^2}\right) \pmod{81}\\ &\equiv 36\frac{f_1 f_6^2}{f_3}U(q) \pmod{81}.\end{aligned}$$

Since $36 \cdot 9$ is divisible by 81, (6.11) proves (6.3) as well. □

Conclusion

The elementary argument in Section 4.3 completes the rank-inequality problem investigated in the original project. It proves positivity in all three progressions $9m + 2$, $9m + 5$, and $9m + 8$, with the common lower bound $p_S(m)$. The essential step is to establish positivity after multiplication by the full partition-generating product; positivity of the numerator alone is neither assumed nor required.

The proof concerns the progression $3n+2$ and uses only the identities already available in the 2018 report. The cubic partition-pair chapters explain known congruences and their proofs, with their original attribution retained.

The accompanying scripts provide exact coefficient checks:

- `verify_completion.py` checks the rank-inequality argument.
- `verify_identities.py` checks the product identities and congruences.

These checks verify formulas and indexing over finite ranges; the proofs and cited theorems establish the results for all indices.

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